REDEM-SSM: A State-Space Architecture with Native Online Learning, Meta-Adaptation, and Structural Plasticity

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Abstract

We instantiate the REDEM mechanisms — an online readout (M1), a slow-trace statistical memory (M3), structural plasticity (M4), and a stability homeostat (M5) — natively on a state-space host (a diagonal linear recurrence with a log-uniform timescale spectrum), instead of retrofitting them onto a frozen-feedforward Transformer. A falsifying development sequence (10-seed paired discipline throughout) isolates three host requirements with mechanism-level evidence. (1) A linear readout on a linearly-decayed state mixture cannot recover the current token exactly (a deconvolution impossibility), and the pooled state readout fails out of sample (0/10 seeds; stream perplexity 58–116 vs. 15.0 for a Transformer+LoRA reference); the readout must use the additive input path Be_{t-1}, which beats the reference 10/10.

Fixed Mamba-style multiplicative gates do not repair the state readout (0/10 seeds at either tested sharpness), so input selectivity must be learned — a design boundary we flag rather than cross. (2) The state's role is statistical metadata, not the readout: a fast-channel state EMA tracks domain switches, and soft routing retains per-domain specialists (forgetting -2.05 to -1.20 ppl, 10/10 seeds at $\tau_m \leq 1000$); the pause-learning (gating-only) policy *improves* stream perplexity on this near-batch RLS readout (10/10), opposite to its effect on the Adam/LoRA readout of the companion study — the policy lesson is readout-dynamics-dependent. (3) Gentle (soft) routing beats abrupt switching (-1.81 ppl, 10/10) and a state-norm homeostat bounds the state and restores the full-state EMA as a valid domain statistic (5/5 switch detections vs. 0.6/5). On a four-domain irregular-switch benchmark

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the full stack beats the bare host (-2.25 stream, -4.47 forgetting, 10/10) and the Transformer+LoRA reference (-9.28, -10.08, 10/10). All results are CPU-scale proofs of concept; we make no scaling claims.

Keywords: online learning, state-space models, continual learning, timescale coverage, homeostatic regulation

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1. Introduction

The REDEM framework couples four adaptive mechanisms: a continuously updated online readout (M1), a slow exponential trace — an EMA of the fast channels — that supplies statistical memory (M3), structural plasticity (M4), and a homeostatic regulator that keeps the substrate at a stable operating point (M5) — the substrate-level regulation of the adaptation process itself is the framework's *meta-adaptation* [2].

A companion study [3] showed that this framework does not instantiate on a frozen-feedforward host: on a Transformer substrate with low-rank adapters, slow-trace domain routing transferred (forgetting -28%, 10/10 seeds) while a gating-only variant was falsified at every metadata timescale (0/10), and the paper concluded that the *full* framework requires a persistent stateful host rather than a retrofit onto a static architecture.

This paper takes the constructive consequence. We design a stateful host — a diagonal linear state-space recurrence with a log-uniform timescale spectrum (the timescale-coverage principle of [3]) — and instantiate M1, M3, M4, and M5 natively. Rather than asserting that the design works, we follow the falsifying discipline of the companion work: every claim is a paired 10-seed comparison with sign consistency, and the negative results are reported as design evidence, not as failures.

The development sequence produced three host requirements that are the paper's main contribution:

1. **The readout needs the input path, not the state** (Section 4.1). Exact recovery of the current token from a linearly-decayed state mixture is a deconvolution impossibility (Proposition 4.1), and the pooled state readout fails in practice (0/10 on a bigram streaming task); the readout must consume the additive input projection Be_{t-1}. Fixed Mamba-style gates do not repair the state readout; selectivity must be learned (a boundary we flag, not cross).

2. **The state's role is statistical metadata** (Section 4.2). A fast-channel state EMA (M3) tracks domain switches and soft routing (M4) retains per-domain specialists; the pause-learning policy is readout-dynamics-dependent.
3. **Gentle wins, and the state must be regulated** (Section 4.3). Soft routing beats abrupt switching; a state-norm homeostat (M5) bounds the state and restores the full-state EMA as a valid domain statistic.

On a four-domain irregular-switch benchmark the full stack beats the bare host and the Transformer+LoRA reference on every seed, and the result transfers to a two-book real-text protocol (Section 4.4). All experiments are CPU-only, tiny by design, and make no scaling claims (Section 6).

2. Related work

State-space models replace the frozen feedforward stack of [11] with a recurrence whose parameters are trained end-to-end ([5, 4, 6, 7]); online adaptation has been pursued by rewriting the state with a learned optimizer ([8]) and by linear-attention delta updates ([9]). Those hosts train with back-propagation, whereas the REDEM line ([1, 2, 3]) restricts the readout to an online, local rule on a fixed substrate, in the spirit of reservoir computing ([13]) and of continual-learning practice ([12]). Our Transformer+LoRA reference ([10], retrained per task as in [3]) is a tiny parameter-efficient adapter; we make no scaling claims against larger models (Section 6).

3. Methods

3.1. Host: diagonal state-space recurrence

The host is a diagonal linear recurrence

$$h_t = A\,h_{t-1} + Be_{t-1}, \qquad A = \mathrm{diag}\left(e^{-1/\tau_i}\right), \tag{1}$$

where e_{t-1} is the one-hot token at step $t-1$, $B \in \mathbb{R}^{N\times V}$ is a fixed random input projection (entries drawn i.i.d. $\mathcal{N}(0, 1/N)$, per-seed), and the timescales τ_i are drawn log-uniformly over $[1, 3000]$ tokens ($N = 128$, $V = 32$). The log-uniform spectrum realizes the timescale-coverage principle of [3]: fast channels carry the recent token, slow channels carry long-horizon statistics. The state is spectrum-whitened,

$$h_t^w = h_t \odot \sqrt{N\,(1 - A^2)}, \tag{2}$$

which normalizes each channel to stationary unit variance (the steady-state scaling of the linear recurrence) and bounds slow-channel magnitude; without it the slow channels accumulate and the RLS covariance grows as $(1/\lambda)^t$ in unexcited directions.

3.2. M1: online RLS readout on the input path

The readout is a per-token recursive-least-squares linear map

$$\hat{y}_t = W\,\phi_t, \qquad \phi_t = [\,Be_{t-1};\ 1\,], \tag{3}$$

with $W \in \mathbb{R}^{V\times(N+1)}$ updated online (predict before update) against the one-hot target e_t with forgetting factor $\lambda = 0.9999$. The cost is $O(F^2)$ per token in the feature dimension $F = N + 1$, not $O(P^2)$ in a parameter count [2, 3]. The squared-loss readout output is a linear MMSE estimate of the conditional distribution, so perplexity is evaluated as $\mathrm{CE} = -\ln \mathrm{clip}(\hat{y}_t[e_t], 10^{-12}, 1)$ *without* a softmax (a softmax double-normalizes the estimate toward uniform). The fraction of clipped (≤ 0) predictions is reported as a diagnostic.

The input-path feature choice is itself an evidence-driven result (Section 4.1); the whitened state h^w is *not* a readout feature.

3.3. M3: fast-channel metadata EMA

M3 is a per-token exponential moving average of the fast channels of the whitened state,

$$m_t = \left(1 - \tfrac{1}{\tau_m}\right) m_{t-1} + \tfrac{1}{\tau_m}\, h_t^w[\text{fast}], \qquad \tau_i \leq 8 \text{ tokens}, \tag{4}$$

with $\tau_m \in \{200, 500, 1000, 2000\}$. The fast channels (which converge in a few tokens) are a stationary domain statistic; the full state is not — its slow channels accumulate over the whole stream, so an EMA of the full state drifts away from any fixed per-domain reference and the detector never flips. Fixed per-domain references are the mean fast-channel state over 1500-token reference streams generated per domain.

3.4. M4: soft routing and dormant-covariance refresh

For D domains, M4 maintains D readouts (W_d, P_d). Routing uses continuous softmax weights over the EMA distances,

$$w_d = \frac{\exp(-\|m_t - r_d\|/\kappa)}{\sum_{d'} \exp(-\|m_t - r_{d'}\|/\kappa)}, \qquad \kappa = 0.5 \times \text{median pairwise } \|r_d - r_{d'}\|, \tag{5}$$

with the prediction $\hat{y}_t = \sum_d w_d W_d \phi_t$ and W_d, P_d updated with error scaled by w_d. This instantiates the "gentle wins" principle of [3] (gradual structural change). A second design detail, *dormant-covariance refresh*, keeps the inverse-covariance P_d of every readout current with the feature stream (weights frozen, P updated with zero error); alone it flips the soft-routing stream result from +1.55 (frozen P) to −3.54 vs. the bare host on the two-domain task.

3.5. M5: state-norm homeostat

M5 is a closed-loop controller on the whitened-state norm: the effective time step is adjusted so that the norm is held in a band around its target $\sqrt{N}$ (the whitening's unit-variance intent),

$$\Delta_t = \mathrm{clip}\big(\Delta_{t-1} + \eta\,(\|h_t^w\| - \sqrt{N}),\ 1,\ \Delta_{\max}\big), \qquad h_t = A^{\Delta_t} h_{t-1} + B e_{t-1}, \quad (6)$$

with $\eta = 0.05$, $\Delta_{\max} = 10$. M5 is a feedback regulator, not a fixed normalization: it responds to the measured state and bounds the slow-channel accumulation that otherwise makes the full-state EMA a non-stationary statistic.

3.6. Tasks and discipline

Two-domain protocol (s18): two biased-bigram Markov domains (shift-1 / shift-7 with disjoint bias sets), alternating every 3000 tokens (known switch instants), 6 segments, 18k tokens. **Four -domain benchmark**: four domains (shifts 1/3/7/11, disjoint 4-symbol bias sets) on a cycling schedule with *irregular* switch intervals (Uniform 2500–3500), 8 segments, ∼24k tokens. All seed rules and metrics (stream perplexity, forgetting perplexity on held-out previous domain, post-switch adaptation time) follow [3] verbatim. Every comparison is paired per seed over 10 seeds with sign consistency ($n_{\mathrm{pos}}/10$); the Transformer+LoRA reference is the tiny online-LoRA model of [3] retrained on each task.

4. Results

4.1. P1/P3a: the readout needs the input path

Fig. 1 shows stream perplexity on the two-domain protocol (10 seeds) for the state-based readout arms and the input-path control, against the Transformer+LoRA reference (TF-A1, 15.01). Every state-feature readout — log-normal spectrum (LN-raw 114.99, LN-whiten 115.87), coverage spectrum

(CV-whiten 81.31), with the current-token projection added (CV-skip 58.47) — is worse than the reference on every seed (0/10). The input-path control ($\phi = [Be_{t-1}; 1]$, no state) reaches 11.75, better than the reference on 10/10 seeds (-3.26). Proposition 4.1 shows why exact recovery is impossible; the corollary below scopes the measured failure.

Proposition 4.1 (Deconvolution impossibility for the linear state readout)**.** *Let the host be $h_t = \Lambda h_{t-1} + Be_{t-1}$ with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$, $0 < \lambda_i < 1$ distinct, and $B \in \mathbb{R}^{N\times V}$ with i.i.d. continuous entries. For almost every such (Λ, B), no affine-linear readout $\hat{e}_t = Uh_t + b$ recovers the most recent input exactly ($\hat{e}_t = e_{t-1}$ for all token histories).*

Proof. Unrolling the recurrence gives the mixture $h_t = \sum_{j\geq 0} \Lambda^j Be_{t-1-j}$ (with $e_{-1} = 0$). Exact recovery for every history requires $UBe_{t-1} = e_{t-1}$ for every token, i.e. $UB = I_V$, and $U\Lambda^j B = 0$ for every $j \geq 1$. The annihilation conditions say U vanishes on $S = \mathrm{span}\{\Lambda^j Bu : j \geq 1,\, u \in \mathbb{R}^V\}$. Because the λ_i are distinct and B is random, the Krylov family $\{\Lambda^j B\}_{j\geq 0}$ generates $\mathbb{R}^N$ almost surely (the N vectors $\Lambda^j b_\ell$ for a single random column b_ℓ form a Vandermonde system with distinct nodes, hence are linearly independent a.s.), so $S = \Lambda\ \mathrm{span}\{\Lambda^j B\}_{j\geq 0} = \Lambda\,\mathbb{R}^N = \mathbb{R}^N$ (Λ is invertible). Hence $U = 0$, contradicting $UB = I_V$. Whitening replaces h_t by Dh_t with a positive diagonal D, i.e. $U' \mapsto U'D$; the argument is unchanged, so the contamination is structural, not a scale artifact. □ □

The empirical corollary of Proposition 4.1 is the measured 0/10 failure of every state-feature readout: a pooled linear readout cannot use the state mixture under the streaming perplexity metric, and the input-path feature $\phi = [Be_{t-1}; 1]$ enters the state equation as the j=0 term — exactly the component the readout must consume. The proposition bounds *exact* recovery; a boundary analysis (data/s35_readout_boundary_probe_v1.csv) shows the measured failure is a property of the pooled readout under this metric rather than an absence of linear information. The current token is linearly decodable from the fast channels ($\tau_i \leq 8$) of the whitened state at 88.9–99.7% out-of-sample accuracy (chance 3.1%; 95.4–99.9% with $\tau_i \leq 30$), whereas the slow channels ($\tau_i \geq 100$) decode at chance (0.034–0.049): the older-token contamination is a slow-channel property, and whitening rescales it without removing it. The in-sample ridge oracle is itself window-dependent (28.8–33.4 on the full window vs. 16.2–18.4 on the first half) and non-monotone in the feature set (adding the clean projection makes it worse on every seed:

16.6–19.9 vs. 7.15–7.36 for the projection alone) — artifacts of evaluating clipped linear outputs by perplexity, not bounds on recoverable information. Restricting the readout to the fast channels does not repair the next-token readout either: fast-channel-only linear fits and two-stage decode-then-table compositions still fail out-of-sample (test ppl 68–104 vs. 13.9–17.4 for the pooled transition table fit on one half and evaluated on the other) — the token is linearly present, but the closed-form squared-loss readout never turns it into calibrated next-token probabilities, which is exactly what the input-path feature achieves by encoding the token itself. Within the RLS-only constraint the honest resolution remains the additive input path.

Fixed Mamba-style multiplicative gates ($\hat{y} = W\,(h^w \odot \sigma(\gamma C B e_{t-1}))$, C fixed random, $\gamma \in \{1, 5\}$) do not repair the state readout: stream perplexity 77.19 (γ=1) / 72.21 (γ=5); the sharper gate does not repair it either (0/10 vs. the reference), so the negative is not a tuning artifact. Selectivity in real Mamba-style hosts is *learned* end-to-end; a fixed random gate carries the previous token only multiplicatively inside a time-varying state, which is noise, not signal. Within our RLS-only constraint the honest resolution is the additive input path — which is what Mamba's input branch provides. We flag the learned-selectivity route as the natural extension rather than crossing the RLS-only boundary in this work.

4.2. P2: routing transfers, gating is readout-dependent

Table 1 and Fig. 2 (left). M3 (fast-channel EMA) tracks all five known switches of the two-domain protocol. Routing (A3) with two domain specialists improves forgetting at $\tau_m \leq 1000$ (-2.05, -1.87, -1.20 ppl; 10/10 seeds), neutral at 2000 (-0.007, 5/10) — the "routing transfers" lesson of [3] holds on the SSM host. A3's stream cost in the frozen-covariance variant (+0.38 to +4.08) is removed by dormant-covariance refresh (Section 3.4, Section 4.3).

The gating-only arm (A2) *improves* stream perplexity at every τ_m (-1.52 to -2.45, 10/10), the opposite of the companion's A2 falsification (0/10). Two differences must be stated honestly: the companion's A2 paused its LoRA *plasticity* while keeping the head *readout* active at every step, whereas here the RLS readout is the only trainable component, so pausing it pauses all learning — the two arms are not the same experiment. The mechanism of the inversion is the readout's dynamics: a near-batch RLS readout ($\lambda = 0.9999$) that pauses within domains stays near the pooled-table optimum, whereas a fast Adam/LoRA readout that pauses lets within-domain knowledge decay. Taken together the two results show that the pause policy's effect depends

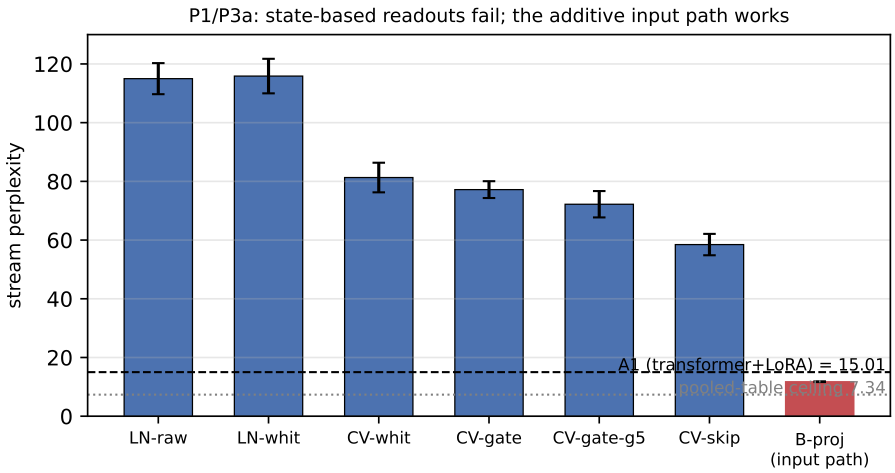

Figure 1: P1/P3a (10 seeds): stream perplexity of state-based readout arms vs. the input-path control on the two-domain streaming task. All state-feature readouts fail (0/10 vs. the Transformer+LoRA reference TF-A1, dashed); fixed gates do not help; the additive input path (Be_{t-1}) reaches 11.75, below TF-A1, approaching the in-sample pooled-table ceiling 7.25 (mean over seeds; range 7.15–7.36, dotted; the same table fit on one half and evaluated on the other transfers at $\sim$15 across the domain-mix shift, so the input path also beats the non-adapting static table).

on the learner's dynamics *and* on what is paused; the companion lesson "the readout must remain active throughout the stream" [3] is not a universal principle. We report this qualification here rather than editing the companion paper.

4.3. P3: gentle wins, and the state must be regulated

Fig. 2 (right): soft routing (M4, Eq. 5) beats abrupt routing on stream perplexity at τ_m=500 (-1.81 ppl, 10/10) while keeping the forgetting gain (-1.99 vs. -2.37 for abrupt — hard switching retains purer specialists). The dormant-covariance refresh alone flips the soft-routing stream result from $+1.55$ (frozen P) to -3.54 vs. the bare host (10/10).

M5 (Eq. 6) bounds the whitened-state norm (mean 11.3 in band vs. 50.2 and growing for the bare host), restores the *full*-state EMA as a valid domain statistic (5/5 switch detections vs. 0.6/5 — the non-stationarity of Section 3.3 fixed at the source), and recovers from a $10\times$ state spike in $\sim$10 tokens (Δ_t settles near 5.4). This is mechanism-level evidence: on this architecture

arm	τ_m	stream	Δ_{ppl}	forget	Δ_{forget}
A1 bare	—	11.75	—	8.39	—
A2 gate	200	10.24	−1.52 (10/10)	8.91	+0.52 (0/10)
A2 gate	500	9.53	−2.23 (10/10)	8.42	+0.03 (5/10)
A2 gate	1000	9.30	−2.45 (10/10)	8.15	−0.23 (10/10)
A2 gate	2000	9.70	−2.05 (10/10)	8.02	−0.37 (9/10)
A3 route	200	12.14	+0.38 (0/10)	6.33	−2.05 (10/10)
A3 route	500	13.30	+1.55 (0/10)	6.52	−1.87 (10/10)
A3 route	1000	14.81	+3.06 (0/10)	7.19	−1.20 (10/10)
A3 route	2000	15.83	+4.08 (0/10)	8.38	−0.01 (5/10)

Table 1: P2 (10 seeds, two-domain protocol): routing (A3) retains domain specialists (forgetting improves at $\tau_m \leq 1000$, 10/10) at a frozen-covariance stream cost that dormant-covariance refresh removes (Section 4.3); gating-only (A2) improves stream perplexity 10/10 at every τ_m — the readout-dynamics dependence.

the readout is decoupled from the state (input path), so a direct "stream performance under disturbance" test is not meaningful here.

4.4. P4: benchmark on four-domain irregular switching

Fig. 3 and Table 2. On the four-domain irregular-switch protocol, the full stack (M1 input-path readout + M3 metadata + M4 soft routing) beats the bare host (stream −2.25, forgetting −4.47; 10/10) and the Transformer+LoRA reference (stream −9.28, forgetting −10.08; 10/10). Even the bare host beats the transformer reference 10/10 (−7.03): the RLS readout's second-order convergence dominates online Adam/LoRA when domains switch faster than the LoRA can track (the reference reaches only 22.46 ppl, below the uniform-chance 32).

Real-text transfer. The same protocol on two public-domain books (Alice's Adventures in Wonderland vs. A Tale of Two Cities, character -level, case-folded, 32-symbol vocabulary) reproduces the result: REDEM-SSM beats SSM-bare (stream −1.27, 10/10; forgetting −0.47, 9/10) and beats TF-A1 (stream −5.23, forgetting −2.01; 10/10); even the bare host beats TF-A1 10/10. The fast-channel metadata separates the two sources, and soft routing retains per-source specialists, although the forgetting gain is smaller than on the synthetic task — the two books share English letter bigram statistics, so their specialists overlap more than the four synthetic domains'. The first-order readout reaches the char-bigram regime (ppl ∼12–13);

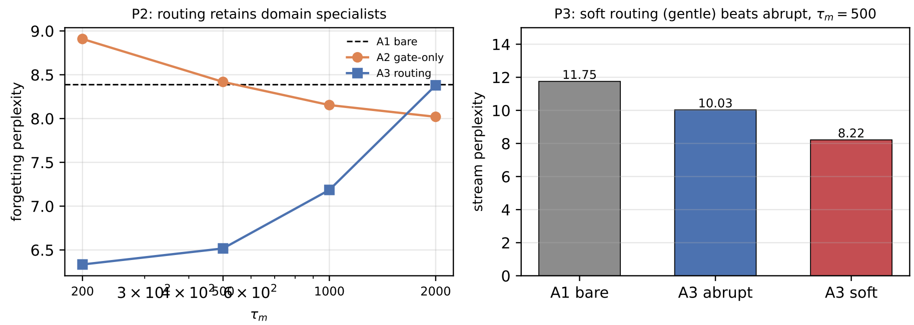

Figure 2: Left (P2): forgetting perplexity vs. τ_m — routing (A3) retains domain specialists below the bare baseline (A1, dashed); gating-only (A2) does not. Right (P3): soft routing beats abrupt on stream perplexity at τ_m=500 ("gentle wins").

higher-order text structure is out of its reach and is reported as a boundary, not a result. A count-based bigram oracle pins this boundary quantitatively (data/s31_char_bigram_oracle_v1.csv): the true per-source first-order ceiling, estimated from the full books and evaluated on the same per-seed windows, is ppl 10.97 ± 0.18 (stream) — REDEM-SSM's 12.07 sits within $\sim$1.1 ppl of its structural ceiling (the residual is finite-sample loss of the streaming readout), and the Transformer reference (17.31) fails to reach even this first-order ceiling, so its deficit is not about higher-order structure.

Reference fairness and mechanism transfer (follow-up). Two legitimate objections to the TF-A1 comparison are that the reference uses the untuned s18 hyperparameters and that it carries no routing mechanism. We close both on the same four-domain protocol (10 seeds; data/s26_ssm_p4_fair_tf_v1.csv). (i) A small tuning grid (four learning rates from 3×10^{-4} to 10^{-2}; ranks 16 and 32) improves the bare reference's stream perplexity to 14.86 at lr=3×10^{-3}, rank 32 — still worse than REDEM-SSM on 10/10 seeds (-1.68, paired $t = 8.8$; the best config's forgetting collapses to 62.2 vs. 8.93, so the tuning that helps the stream destroys retention and the default reference is the balanced representative). The P4 margin is therefore robust to reasonable reference tuning, though smaller on stream. (ii) The s18 routing mechanism (A3) generalized to four adapters (TF-A3, τ_m=500) recovers the specialists: forgetting drops from 19.01 to 10.84 (-8.17, 10/10) — slow-trace routing transfers to the Transformer host on this second protocol (Paper C) — but does not repair the stream: TF-A3 stays at 21.77, losing to REDEM-SSM on 10/10 seeds (+8.59). The stream

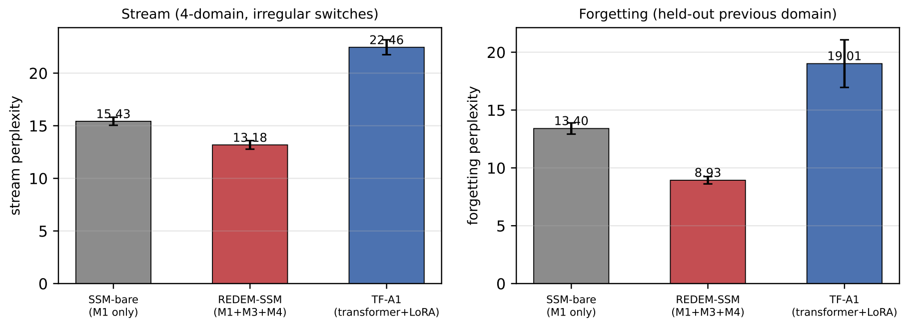

Figure 3: P4 (10 seeds, four-domain irregular-switch protocol): the full REDEM-SSM stack beats the bare host and the Transformer+LoRA reference on both stream and forgetting perplexity (paired, 10/10).

arm	stream ppl	forgetting ppl
SSM-bare (M1 only)	15.43	13.40
REDEM-SSM (M1+M3+M4)	**13.18**	**8.93**
TF-A1 (transformer+LoRA)	22.46	19.01
paired diffs (10/10): REDEM–bare −2.25/−4.47; REDEM–TF −9.28/−10.08; bare–TF −7.03/−5.61		

Table 2: P4 (10 seeds, four-domain irregular-switch protocol): the full stack (input-path RLS readout, fast-channel metadata, soft routing over four specialists) beats the bare host and the Transformer+LoRA reference on every seed. The transformer reference uses the companion's hyperparameters, untuned for four domains.

advantage of the input-path RLS readout is a property of the stateful host, not of the routing mechanism: the mechanisms are host-agnostic, the stream performance is not.

M5 in the benchmark (negative). The P3 state-norm homeostat was deliberately excluded from the P4 stack; adding it confirms the decision empirically (data/s33_ssm_p4_m5_v1.csv): REDEM-SSM+M5 is *worse* on all 10 seeds (stream +1.53, forgetting +2.90; paired $t = -22.8/-26.5$). The mechanism is structural: the Δ_t modulation departs from the per-channel whitening calibrated at $\Delta_t = 1$, so the readout's features lose their calibrated scale on a stream that is already well-conditioned (the norm never runs away — mean $\Delta_t \approx 5.3$ means the controller is constantly active). M5 remains a mechanism-level safeguard for the spike regime (P3, E2), not a benchmark component.

5. Discussion

Three evidence-based host requirements emerged. **(i)** A linear readout cannot use a linearly-decayed state mixture under the streaming perplexity metric: exact recovery is a deconvolution impossibility (Proposition 4.1), and the pooled readout fails in practice, so the current token must be an additive input feature (Section 4.1); fixed gates do not substitute for learned selectivity. **(ii)** The state's role is statistical metadata: its fast channels, not the readout, carry the domain statistic (Section 4.2); on first-order tasks the slow channels are inert for the readout, and their accumulation must be masked (Section 3.3) or regulated (M5, Section 4.3). **(iii)** The policy lessons of [3] transfer in part: routing retains specialists, gentle structural change beats abrupt, but the pause-learning policy's effect is readout-dynamics-dependent (10/10 improvement here vs. 0/10 there), so "the readout must remain active" is not a universal principle.

The honest negative results are the paper's spine: the naive state readout, the fixed gate, the full-state EMA metadata, and the frozen-covariance routing all failed in ways that *located* the working design (input path, fast-channel metadata, soft routing, dormant-covariance refresh, state regulation) rather than merely being abandoned.

6. Limitations

CPU-only, toy-scale (state 128, $\sim$4k readout parameters); the input-path readout is a first-order bigram model — the real-text benchmark transfers (Section 4.4) but higher-order text structure is out of reach; linear host (no chaos; M5 is a stability-edge regulator, an analogy to the homeostat of [1, 2], not a validation of it); the learned selectivity route (real Mamba-style gating) is not crossed; the transformer reference uses the companion's hyperparameters untuned for four domains; hyperparameters (τ_m, κ, λ, η, $\Delta_{\max}$) are fixed, not swept. No scaling claims.

7. Conclusion

REDEM mechanisms can be instantiated natively on a state-space host, and the falsifying development sequence locates the working design precisely: an input-path RLS readout, fast-channel statistical metadata, soft routing with dormant-covariance refresh, and a state-norm homeostat. The full stack beats the bare host and the Transformer+LoRA reference on a four-domain irregular-switch benchmark on every seed. The next steps are real-text evaluation, learned selectivity, and scaling — each with the same falsification discipline.

Code and data availability

All simulation scripts and generated data are available at `https://github.com/huyamingc/REDEM` (private during review; released on acceptance).

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